

The official website for this project is <https://sourceforge.net/projects/iiot-gateway/>. It has a mailing list, bug reports and 'Help' documents for users.

Applications can run locally in factory premises

When asked why he thought of developing something like this, Joshi says that existing hardware and software solutions are very expensive and require costly training. They also have strict licence requirements that are hard for new developers to meet. Moreover, emergency support for customised applications is quite difficult to get and because of the proprietary nature of the applications and design, one cannot use Google or the Web to search for answers like IT people normally can do. It is difficult to find help for any issue in industrial applications. This leads to financial losses for end users and in turn, a loss of reputation for the systems integrator. Joshi adds that the technology used in the Indian manufacturing and production industry is quite old, and none of it is taught to students in colleges or training institutes.

Joshi says, "After spending many daunting years in this field, one fine day in late 2018, I decided that enough was enough — unless someone designed a system made for India which would enable affordable and simplified application development (designing while making use of open technology), I would continue facing issues on and off-field. And given my core competency and interest in Linux, I decided that someone ought to be me. So I started development and testing applications from mid-2019. There was a special emphasis on ensuring that the open standards operational in the industry were supported out-of-the-box."

This OS can be used to design, deploy and develop applications for machine monitoring, data acquisition, and visualisation in core industries like the automotive, pharma and process

sectors. Unlike other existing systems, the OS is designed keeping network administration and remote users in mind. The applications can be hosted on the cloud, and users can get or view remote data on their mobiles using an Android application. Apps can run locally in factory premises or can be hosted on the Amazon Web Service (AWS) cloud platform.

No more expensive and proprietary software or technology

What makes IIoT Gateway unique? Joshi says that it is the first OS distro that's been made in India, which focuses on the Industrial Internet of Things — and it's free for use. No existing player in the IT or industrial sector has done something like this before. "The user is empowered with technology that can be used as per requirements, with no hassles involved. It also has inbuilt application designing facilities in high level languages like Python, NodeJS, and Java. Computer science graduates, tech enthusiasts, and people interested in making career shifts can try to cross over considering the impending slowdown in the Indian IT sector. There needs to be no more relying on expensive and proprietary software or technology to get a head start in your career — everything is open. A DIY person will find this better," Joshi says.

So, why did Joshi open source this innovation? He replies, "The base of IIoT Gateway is the Linux kernel, which is the heart of the overall OS. It is released to the public under the GNU License, which requires users to release all the source code they have developed to the public; so any work that I or anyone does is required to be released under the Open Source Software Clause at no or minimum maintenance cost. This is applicable for all base software like Python, NodeJS and Java that are used to develop applications. However, the code that is enterprise-specific or which has got some trade secret, cannot be released to others, since every user's

requirements differ. For that, we can have a different licence like LGPL."

The benefits of IIoT Gateway

Joshi states that the open source model comes with various benefits like flexibility, agility, cost effectiveness, information security and bug fixes. He adds that open source enables technology agility, typically offering multiple ways to solve problems. It helps prevent IT organisations from getting bogged down just because a particular capability isn't available from a vendor. Instead of waiting for the vendor to deliver that capability, it can be created inhouse. Also, open source can reduce the planned obsolescence, which costs enterprises a lot when working with closed source developers.

Open source solutions are also typically much more inexpensive, and they give enterprises the ability to start small and scale up.

Joshi adds, "Open source has a solid information security record. The responsiveness of the open source community and vendors with respect to information security problems has been very good."

Enterprise problems can be solved while effectively sharing some of the maintenance costs. One big advantage of open source is the community involvement. Instead of writing an application and having to sustain it yourself, one can share the cost of maintaining and sustaining applications among multiple participants from the community. Joshi adds, "Lastly, building a community around open technology is more beneficial than maintaining closed source software for personal benefits."

This gateway can be accessed at https://sourceforge.net/projects/iiot-gateway/files/20_may_2020/installer_19052020.iso/download. It will download a 1.5GB DVD image, which can be burnt and run on a real computer or any virtual machine software. Instructions and